

Guide to NPS Water Balance Repository

Madeleine Burns
Scientists in Parks Program
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Introduction

The NPS Water Balance repository on GitHub is the publicly available, centralized location of the National Park Service (NPS) water balance and streamflow models. The repository, available [here](#), contains scripts to calibrate and run both models for historical and future time periods, well as conduct a variety of statistical analyses. This document describes the theoretical basis of the model and provides a practical guide for accessing and running the code in the repository.

Theoretical Overview

Water Balance

The NPS water balance model is based on the 1955 work of Thornthwaite and Mather, with continued modifications and development by David Thoma, Mike Tercek, and other members of the NPS Inventory and Monitoring Division and Climate Change Response Program (M. Tercek et al., 2018; Thornthwaite & Mather, 1955). Water balance is the mathematical accounting of water input and movement through the environment. It accounts for water storage in different phases (ice, liquid, gas) and movement between stages via processes including evapotranspiration or melting. This version of the water balance model accounts for heat load on different aspects and slopes in complex terrain, water holding capacity of different soil types, and local site characteristics such as slope, aspect, and soil properties. This allows the model to accurately depict biophysical conditions at local scales that plants and animals experience. Water balance variables such as evapotranspiration or water deficit are usually more strongly correlated with natural resource response (i.e. plant growth, fire danger, or habitat extent) than temperature or precipitation, so the water balance model provides a valuable tool for understanding the impacts of water availability on natural resources and project future changes in parks and in other locations (Stephenson, 1998).

The water balance model has been used for a range of NPS projects and analyses, including evaluation and identification of wetland drying, changes in plant water use, pivot points in drought periods, and wildfire ignition (Ray et al., 2019; M. T. Tercek et al., 2021; D. P. Thoma et al., 2019, 2020). The NPS has run a gridded version of the water balance model for the entire United States from 1980-2100 at a scale of 4km for historical projections and 1km for future projections. The gridded water balance projections can be found [here](#) (the products are not always be publicly accessible due to server costs; if the link isn't working, contact miketercek@yahoo.com for access). However, the spatial resolution of the gridded water balance product may not be representative of site-specific conditions, especially in areas of high relief or high topographic complexity and high variation in soil water holding capacity. The code in this repository allows the water balance model to be run for a specific point location or watershed area with user-defined parameters, increasing the granularity and applicability of the model. Additionally, the repository includes a streamflow model that combines the water balance model with an IHACRES flow model, enabling streamflow estimation and broadening the code's applicability to key hydrologic questions related to surface water.

The full water balance calculations include many intermediate variables; when the model is run, these variables are saved and exported into CSV files. Six of the intermediary water balance variables were selected to be plotted and analyzed in more depth in this code because of their relevance to hydrological and ecological analyses. These variables, discussed in more detail below, are soil water content (`soil_water`), runoff, snow water equivalent (`accumswe`), potential evapotranspiration (PET), deficit, and actual evapotranspiration (AET).

Soil Water Content

Soil water content is the amount of water in the top meter of the soil profile. Soil water content influences water availability for plant use, transportation of nutrients through the soil, and hydrological processes such as infiltration, runoff, and groundwater (Haj-Amor et al., 2023). Quantifying the amount of water in the soil is an important way to assess current climatic conditions and project future climate change impact ranging from droughts to floods (Haj-Amor et al., 2023). In the NPS water balance model, the soil water content begins at the same value as the soil water holding capacity (the maximum possible soil water

Commented [BMC1]: Lots of this was taken from Janelle code + Connor report and modified for relevance here

content, a value that is calibrated during optimization) and is iteratively changed at a daily time step as the model is run depending on the amount of available water at the soil surface.

Runoff

Runoff is water that falls as precipitation and moves across the landscape toward streams rather than infiltrating into the ground. Although runoff contributes substantially to streamflow, it does not account for the entire volume of streamflow, as groundwater also supplies water to streams. In this repository, the IHACRES model accounts for the influx of groundwater and converts runoff from the water balance model to streamflow. Runoff efficiency, or the ratio of runoff to precipitation, quantifies how much water will be available as runoff and serves as an indicator of the hydrologic dynamics of an ecosystem. In the NPS water balance model, runoff is modeled as the available water from precipitation, minus water loss from soil infiltration and evapotranspiration.

Snow Water Equivalent

Snow water equivalent (SWE) is the amount of liquid water contained within the snowpack. In snowmelt-dominated regions, SWE provides an important measurement of water during the winter months, used to estimate water availability and forecast flood risk during later months. SWE is also an important metric for assessing long-term climatic change within the cryosphere. In the NPS water balance model, SWE is modeled as the combination of snowfall (the solid fraction of precipitation) and snowmelt.

Potential Evapotranspiration

Potential evapotranspiration (PET) is the maximum amount of water that could be evaporated from the landscape (including surface water, plants, and soil), assuming unlimited water availability. PET measures water demand and is an important metric for evaluating trends in evaporative demand due to climate change. In the NPS water balance model, PET can be calculated using the Penman-Monteith, Hamon, or Oudin methods (Allen, 1998; Hamon, 1963; Oudin et al., 2005). The below figure shows comparisons of the three methods for Little River, TN, with observational data from OpenET (purple) as a reference point.

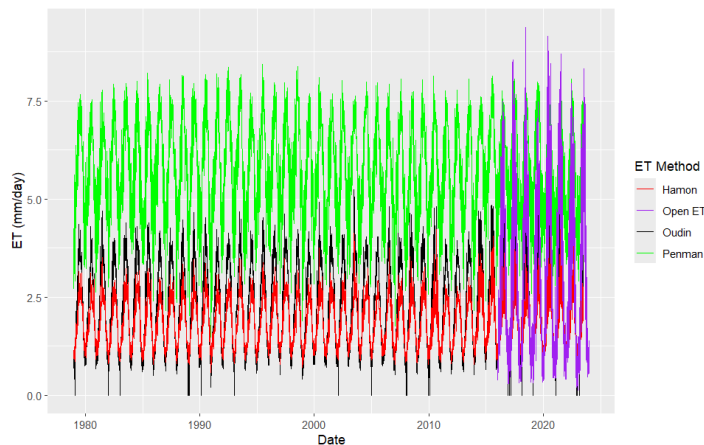


Figure 1. Comparison of methods for calculating PET with estimates of AET from OpenET. Penman consistently produces higher AET estimates, likely because Penman explicitly models solar radiation and wind in more detail than either of the other methods.

Actual Evapotranspiration

Actual evapotranspiration (AET) measures how much water was actually evaporated from the landscape, taking into account water availability. AET provides an accurate summary of how much water is lost through evaporation from soil and surface water, as well as through plant transpiration. Similarly to PET, it provides a useful metric for evaluating evaporative demand and atmospheric water storage, as well as water availability. AET is most similar to PET in locations with significant available water (such as the southeast United States); in contrast, AET will differ significantly from PET in arid regions. Comparing AET to PET can indicate whether a region is water-limited ($AET < PET$) or energy-limited ($AET = PET$). In the NPS water balance model, AET is calculated from the available water from precipitation and the soil water content; the value of AET is capped so that it does not exceed the calculated PET.

Deficit

Deficit (D) is an estimate of unmet water need, calculated as the difference between PET and AET. Because deficit represents the amount of water vegetation would use if the water was available, it serves as an estimate of vegetation drought stress and is useful for understanding vegetation health and ecosystem dynamics. In the NPS water balance model, the deficit is modeled as the difference between PET and AET.

Together, AET and D are good predictors of vegetation condition. Many species have relatively narrow climatic windows (intersections of AET and deficit) in which most members of the species can survive. These climatic windows are useful for defining ecological niches, as well as species distributions within ecosystems. Evaluating changes to AET and deficit in future conditions is a useful method for understanding climate stress on vegetation and future transitions in vegetation species or distribution (Stephenson, 1998). If the clustering of AET and D shift outside of the historical climatic window, vegetation assemblages will change and adapt to the new climate space.

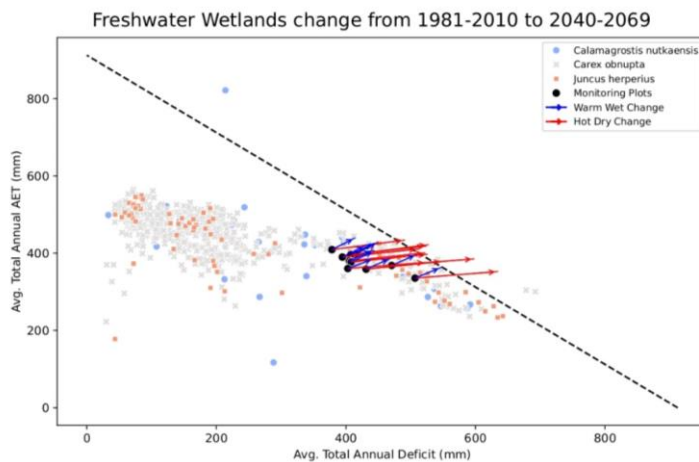


Figure 2. Annual total AET and deficit are plotted for freshwater wetlands; black dots indicate freshwater monitoring plots in the NPS San Francisco Bay Area Inventory and Monitoring Network (SFAN) and

arrows indicate changes under projected climate futures. In the Hot Dry future, the SFAN plots will move out of the climatic window, indicating that a change in species distribution and abundance is likely.

Streamflow

The streamflow model includes two steps: first, a water balance model generates estimations of runoff using meteorological forcings data and second, a runoff model uses these estimations of runoff to model streamflow. The water balance component is the NPS model described in the above section. The runoff component is based on the IHACRES methodology, which models streamflow as a combination of unit hydrographs with flow components of different speed (quick, slow, and very slow) (Littlewood, 2021). The combined NPS water balance – IHACRES flow model was developed more recently and has been used in climate change analyses of streams in Arizona and California (Burns et al., 2025; D. Thoma et al., 2025). Overall, the NPS model has proven to be a reliable model that can be accurately calibrated to a variety of types of watersheds and performs similarly to well-established models (Stern et al., 2025; D. Thoma et al., 2025).

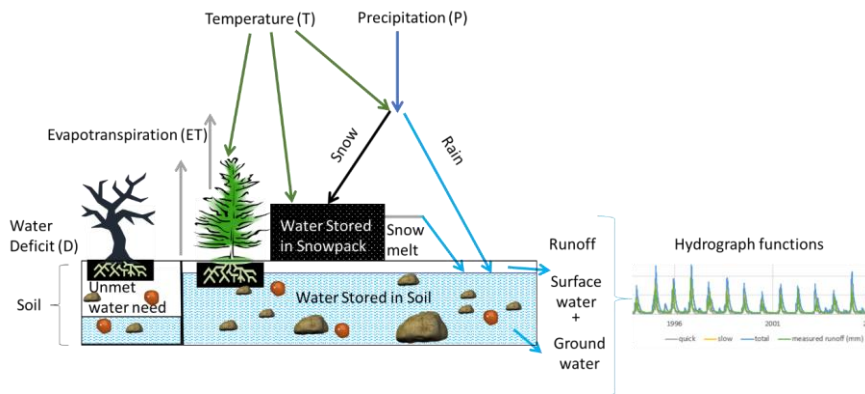


Figure 3. Visual representation of the streamflow model, which is comprised of a water balance model paired with a rainfall-runoff model, modified from McCabe & Markstrom, 2007.

Input Data

Both the water balance and combined water balance-streamflow models require meteorological forcing data as input. The models can be run for historical time periods, using observational meteorological data or gridded reanalysis products as input, or for future time periods, using future meteorological projections from climate models as input. The code in this repository currently supports the use of GridMET meteorological data (Abatzoglou, 2013), Daymet meteorological data (Thornton et al., 1997, 2021), or MACA-downscaled CMIP 5 projections for the historical time period and MACA-downscaled CMIP5 projections (Abatzoglou & Brown, 2012) for the future time period. GridMET data is accessible from 1979 to yesterday at a 4km gridded resolution across the contiguous United States. Daymet data is accessible from 1980 to the present day at a 1km gridded resolution across North America and Hawaii. Both datasets are daily resolution. Historical MACA projections are available from 1960-2006 and future MACA projections are available from 2005-2100, both at a 4km resolution across the contiguous United States. Out of the 20 CMIP5 models that MACA provides downscaled data for, the code currently pulls RCP4.5 and RCP8.5 scenarios for 13 models, generating an ensemble of 26 future scenarios. Those models are BNU-ESM, CCSM4, CNRM-CM5, CSIRO-Mk3-6-0, CanESM2, GFDL-ESM2G,

HadGEM2-CC365, IPSL-CM5A-LR, MIROC5, MIROC-ESM-CHEM, MRI-CGCM3, NorESM1-M, and Inmcm4. Users can identify additional or different GCMs to use when running the model, if desired.

Though the model is calibrated using GridMET data, MACA projections are used as historical data whenever future projections are compared to historical data. The bias correction methods used in the MACA downscaling process introduce spatial and temporal discrepancies between GridMET historical data and MACA future projections, which can cause inaccurate comparisons of those datasets (i.e. the comparison might suggest a decline in precipitation, but this decline is simply a result of mismatched bias correction/interpolation methods). Therefore, to ensure that the comparison is as accurate as possible, MACA historical data is used when comparing future projections to the historical time period. Because future projections from one model are compared to historical projections from the same model, variance due to the specific model and downscaling techniques are eliminated, leaving just the climate signal.

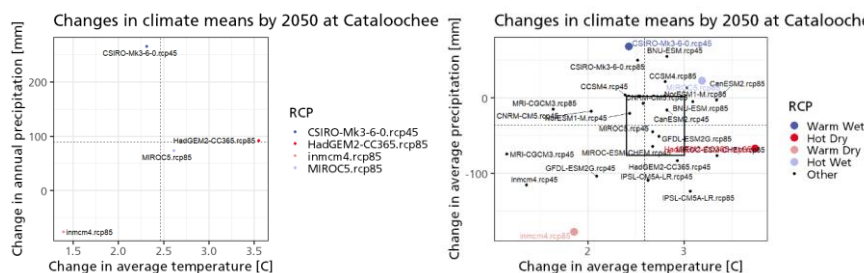


Figure 4. Changes in annual climate means at Cataloochee River. The plot on the left compares historical MACA projections to future MACA projections for each model, while the plot on the right compares historical GridMET projections to future MACA projections. Changing the historical data results in a significant difference in the change in annual precipitation (the MACA comparison shows a large increase for three models, while the GridMET comparison shows a large decrease for two models and a very small increase for a third model). This is likely because the selected watershed is located in a topographically complex region where GridMET interpolations and MACA downscaling did not align.

Meteorological forcing data can be pulled either for a point location or averaged over the entire area of the watershed; this choice is defined in the code as explained in the Input Data section. Area averages smooth out extremes, especially in precipitation, which may result in less accurate results for areas like the western U.S. that often have small-scale, flashy precipitation events. Area averages are also less likely to capture small-scale variation in topography, making them less accurate for mountainous terrain. Ultimately, area averages will produce more accurate results for large watersheds where cumulative precipitation across a large spatial area influences streamflow, and point locations should be selected when the scale of analysis is small or the watershed includes significant topographical variation or variation in meteorological patterns.

Pulling data is often the most time-consuming component of running this model and requires a strong Internet connection. Additionally, the ClimateR package ([here](#)) that is used to access meteorological data has, in the past, changed its functions or return values without warning. If meteorological data pulled using functions from this repository doesn't appear to be correct, it is likely due to a change in the ClimateR package. I have consistently had more luck using the Python version of the package, ClimatePy, to accurately and quickly pull climate data. The Code folder contains a Jupyter notebook titled

`get_climate_data.ipynb` that contains code to pull MACA and GridMET data for historical and future time periods, given a user-provided shapefile.

Repository Overview

The repository has five folders: Water Balance, Streamflow, Code, and Validation. The Water Balance and Streamflow folders contain scripts to run the water balance or streamflow models (respectively) for either point locations or area averages in the United States. These folders also contain supporting scripts that calibrate the model, assess its accuracy, and analyze historical and future model projections. The Code folder contains supporting scripts that define functions used by both models. The Validation folder includes scripts to validate the R version of the model using the original Excel version of the model, as well as the more recently developed Python version of the model. The Data folder initially contains two files with information necessary for the code to run; as the code is run, additional intermediate data (such as climate projections or USGS stream gage data) is stored in the Data folder as CSV files. When the code is run, an additional Output folder is created to store figures and model parameters that are generated by either model. The default spatial domain of both models is a watershed with a USGS stream gage, defined by the user, though the user can provide coordinates of a point location and/or a shapefile if they want to define their own spatial domain.

Within each folder, sub-folders are created for each location that is modeled, according to the SiteID that the user names. Within the Output folders, additional sub-folders are created to distinguish between different model runs; these are also named by the user with the variable FolderName.

All “Run” scripts, including `Run_WB_Model.R` and `Run_Streamflow_Model.R` and the multi-watershed and multi-parameter versions of each of those scripts, are designed to be as simple to run as possible. Before running the script, the user adjusts the parameters at the beginning, specifying the location and the model configuration that they want to run. Then, the user runs the script. Though various sub-scripts are called during the running process, these sub-scripts are designed to interface with the main script; the user does not need to manually run each script (though they can edit the sub-scripts if they are interested in new types of analyses or plots). Additionally, each of the models described here are currently in “runnable” form; in other words, each script has a valid site name and USGS gage number that has previously been tested on the model. New users can simply fork a version of the code onto their device and run the pre-existing scripts if they want to get a sense for how the model works.

Below, detailed descriptions of each user-specified parameter for both models are included. The two models share many common parameters, particularly those related to configurable run options. The flowchart below offers a concise overview of these parameter choices and their influence on model setup, helping users explore different configurations and understand how they affect model accuracy.

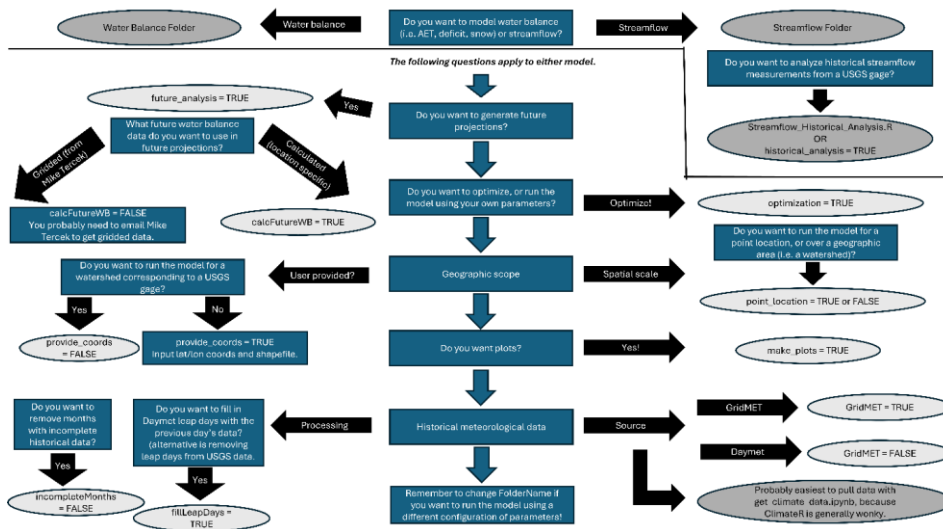


Figure 5. Users can specify the temporal and spatial resolution of the model, the type of parameter used in the model (optimized vs. user-input), whether plots are generated, and the source and data processing of the historical meteorological data. Dark grey ovals indicate files or scripts and light grey ovals indicate parameters within those scripts that can be adjusted.

Water Balance Scripts

The Water Balance folder includes scripts to run a water balance model using user-provided parameters, with an optional step of calibrating the model to OpenET estimates of AET. Because OpenET is only available in 17 western U.S. states (Arizona, California, Colorado, Idaho, Kansas, Montana, Nebraska, Nevada, New Mexico, North Dakota, Oklahoma, Oregon, South Dakota, Texas, Utah, Washington, and Wyoming), the optimization/calibration step can only occur for locations in those states. For other locations, the user must provide parameters for the water balance model.

Run_WB_Model.R

The script has four parts: defining model parameters and setting initial values of variables, pulling OpenET observations and meteorological forcing data, running the model (including optimization, if desired), and analyzing model results. Each of these parts are carried out by sub-scripts called in the main script, described in more detail below.

User Input

The top of the `Run_WB_Model.R` script includes a section that requires user input, titled “Set user-defined variables.” A range of parameters require user input for the model to run:

General variables:

- SiteID: name of location for analysis, used to make plots and folders to store output. Keep it simple. For example, “Redwood Creek.”
- GageSiteID: USGS gage number. [This website](#) provides guidance for finding the gage number on the USGS [StreamStats](#) map.
- FolderName: name of folder created to store output from this run (including plots and results files).
- optimization: TRUE to run optimization code, replacing user-defined model parameters with optimal parameters. FALSE to run model with user-defined parameters. Regardless of which optimization option is selected, the model will read in parameters from an optim_results.rds file if it exists (so a new folder should be created, to run the code with user-provided input variables).
- optimization_var: ‘AET’ when optimization of water balance parameters is taking place and/or when the water balance code is being run with calibration coefficients for AET.
- future_analysis: TRUE runs the `Future_Analysis.R` script to generate and analyze future streamflow projections.
- calcFutureWB: TRUE to re-run entire water balance model for future; FALSE to use pre-existing water balance projections from a Mike Tercek spreadsheet. If Mike Tercek spreadsheet doesn’t exist in Data folder, this must be TRUE for future_analysis to work.
- make_plots: TRUE generates plots that are stored in the Output folder. FALSE does not.
- provide_coords: if TRUE, user provides coordinates for a point location or area of interest (shapefile). if FALSE, the area of interest is defined as the watershed and point location coordinates are pulled from centroid of watershed using the user-provided USGS gage number.
- point_location: if TRUE, the model is run for a point location; if FALSE, the model is run as an area average.
- percent_skill_cutoff: value between 0-1 that defines what percentage of GCMs are included in the future analysis (models are ranked by skill and selected). Generally, 0.1 is used, meaning that 90% of the models are included (Runyon et al., 2024).

Meteorological data variables:

- incompleteMonths: FALSE to remove incomplete months from monthly comparison of measured vs modeled flow, TRUE to include incomplete months. Set to TRUE for comparison with Excel.
- GridMET: TRUE causes model to use GridMET climate data, FALSE causes model to use Daymet climate data.
- fillLeapDays: only applicable to Daymet data (which does not contain leap days; GridMET and USGS gage data do contain leap days). TRUE fills in missing Daymet leap days with previous days' data. FALSE removes leap days from USGS streamflow data.
- startY, startM, startD: start year, month, and date, in the form 2025-01-01. The GridMET dataset begins on 1979-01-01 and the Daymet dataset begins on 1980-01-01, so these are the recommended respective start dates for historical calibration. This date does not need to match the start date of the USGS gage data; this is automatically adjusted in the code.
- endY, endM, endD: end year, month, and date, in the form 2025-01-01. The GridMET dataset extends until yesterday and the Daymet dataset extends until the end of the most recent calendar year.

Water balance variables:

- PETMethod: options are Oudin, Penman, and Hamon. The default method is Oudin, which is used in the gridded water balance product and generally produces more consistently accurate results than Penman or Hamon, especially for hydrological applications.
- userSetJTemp: TRUE for the user-defined Jennings coefficient and FALSE for it to be automatically extracted from the Jennings TIFF based on the latitude and longitude of the watershed location. Typically set to FALSE

```

22 ~ #####
23 ### Set user-defined variables ###
24 PETMethod = "Oudin"
25 optimization = TRUE
26 optimization_var = 'AET'
27 delaystart = TRUE
28 incompleteMonths = FALSE
29 GridMET = TRUE
30 fillLeapdays = TRUE
31 future_analysis = TRUE
32 calcFutureWB = FALSE
33 userSetJTemp = FALSE
34 make_plots = TRUE
35 provide_coords = FALSE
36 point_location = FALSE
37 percent_skill_cutoff = 0.1
38 FolderName = 'optim'
39 filename = ''
40
41 ### Define watershed ###
42 # must be west of approximately -90 longitude to have OpenET data
43 # USGS gage number: https://waterdata.usgs.gov/nwis/rt
44 SiteID = "Redwood Creek"; SiteID_Filename = gsub(pattern = " ", x = SiteID, replacement = "")
45 GageSiteID <- '11460151'
46
47 ### Provide geographic data or pull shapefile from StreamStats database ###
48 if(provide_coords) {
49   lat = 37.9; lon = -122.59; aoi <- st_read("") # add path to shapefile
50 } else{
51   coords <- get_coords(SiteID_Filename, GageSiteID); lat <- coords$lat; lon <- coords$lon; elev <- coords$elev; aoi <- coords$geometry
52 }
53 region <- get_region(lat,lon)
54
55 ### Define time period for historical analysis ###
56 starty = 2000; startM = 01; startD = 01
57 endy = 2024; endM = 12; endD = 31

```

Figure 6. Variables that require user input are defined at the top of the Run_WB_Model.R script.

Optional parameters:

- Scaling factors for GridMET climate data (tmmx_slope, tmmx_bias, tmmn_slope, tmmn_bias, p_slope, p_bias): to address known inaccuracies in GridMET data, a linear regression model fit to a daily time step and compared to weather station data can be used to determine scaling coefficients for temperature and precipitation. If this analysis has not been carried out, set slopes to 1 and biases to 0.
- Default model parameters: default values for all model parameters are set after the required model parameters. If a file containing optimized coefficients exists from a previous run, those values are set as the defaults; otherwise, the user must define the parameter values. If optimization is TRUE, default parameter values do not matter, as they will be replaced with optimal parameters. Soil.Init, Snowpack.Init, and T.Base are not included in the optimization.
- Model parameter optimization limits: upper and lower limits for model parameter optimization. These values were selected to span the range of physically plausible values for each parameter; therefore, changing the values is not recommended.
- gcm_list: provide list of model (GCM) names in the format 'BNU-ESM.' Downscaled versions of these models will be pulled and used in the analysis of future conditions.

- Latitude and longitude and AOI: if provide_coords is TRUE, provide latitude and longitude and/or a shapefile corresponding to the location that climate data will be pulled for. This is most often the centroid of the watershed (as opposed to the location of the stream gage, which may not be representative of conditions across the entire watershed).

```

65 ##### Set model variables #####
66 ## Set model variables ##
67
68 # Default water balance variables
69 gw_add=0; vfe = 0.755; jtemp = 1.902941; jrange = 3; hock = 4; hockros = 8;
70 dro = 0; mondro = 0; aspect = 180; slope= 0; shade.coeff= 1; SWC.Max = 200
71 k_c = 1; et_slope = 1; et_bias = 0
72 soil_init = SWC.Max; snowpack_init = 0; T_base = 0
73
74 # Get jtemp
75 if (userSetJtemp) {
76   j.raster = raster(here("data", "merged_jtemp.tif"))
77   jtemp = get_jtemp(lat = lat, lon = lon, j.raster = j.raster)
78 }
79 # Water balance optimization lower and upper limits
80 wB_lower = c(gw_add=0, vfe = 0.25, jrange = 1, hock = 0.25, hockros = 0.25, dro=0, mondro = 0, aspect=0, slope = 0, shade.coeff = 0.1, SWC.Max = 10, jtemp = jtemp-0.5, k_c=0, et_slope=1, et_bias=5)
81 wB_upper = c(gw_add = 1, vfe = 1, jrange = 5, hock = 8, hockros = 8, dro = 1, mondro = 1, aspect = 180, slope = 90, shade.coeff = 1, SWC.Max = 400, jtemp = jtemp+0.5, k_c=5, et_slope=1, et_bias=0)
82
83
84 ##### Set other variables #####
85 ## Set other variables ##
86 # optional scaling factors for GRIQNET time series: if no scaling, set slopes to 1 and bias to 0
87 tmm_slope = 1; tmm_bias = 0
88 tmm_slope = 1; tmm_bias = 0
89 p_slope = 1; p_bias = 0

```

Figure 7. Users can choose to define default model parameters and lower and upper limits for optimization.

WB_Optimization.R

The optimization of the water balance model compares modeled AET to OpenET estimates of AET on a monthly time step, using the monthly NSE as the metric for accuracy (Melton et al., 2022). This script is intended to run in conjunction with `Run_WB_Model.R` and cannot be run independently. AET was selected as a parameter for optimization, rather than PET, because OpenET only estimates PET for a grass reference crop, making it an inaccurate reference for ecosystems with more complex vegetation distributions. Though AET is based on satellite measurements and is thus more representative of real-world conditions than PET, it is important to note that OpenET estimates of AET are modeled, though they use observational satellite data as input (Melton et al., 2022). Given the dearth of accurate evapotranspiration data, however, OpenET data represents the most widespread and easily available source of historical evapotranspiration data that could be used to calibrate the water balance model. Improving the data availability and optimization of the water balance model are major areas for potential future development.

The calibration to AET data includes an additional 3 parameters: the crop coefficient (k_c), the AET slope (et_slope), and the AET bias (et_bias). The crop coefficient is used to adjust the magnitude of the calculated PET. The AET slope and bias are used to adjust the magnitude and distribution of the calculated AET, with the constraint that AET must always be less than PET. When the optimization was conducted using just the NSE as the metric of success, this constraint frequently resulted in AET equaling PET for nearly every day in the time series. To prevent this, a penalty was added to the optimization metric, creating a new optimization metric: $NSE - \text{penalty}$. Currently, the penalty involves subtracting 0.27 from the mean number of days where AET equals PET; 0.27 was the fraction of days where AET equaled PET in the Redwood Creek watershed, based on gridded water balance data for the historical period. When applying the model to a new watershed, it is recommended to identify the fraction of days where AET equaled PET using the gridded water balance product, and replace 0.27 with that fraction, to ensure that the optimization function accurately represents the physical characteristics of a given watershed. Ideally, a optimization function could be found that doesn't require manual entry of this percentage, but it has proven challenging to identify while retaining physical realism in the structure of the water balance model. Continued development of the AET calibration is suggested before it is used in analyses.

The optimization function used for this process is a genetic algorithm, from the R package GA. Genetic algorithms (GAs) are stochastic search algorithms that mimic the processes of evolution and natural selection to determine the optimal solution. Specifically, the GA optimization function was selected for this purpose because it allows the user to place physically relevant constraints on parameter values and it searches the entire parameter space, allowing for a basic sensitivity analysis of the model. Additionally, a GA optimization also runs more quickly than a manual optimization. The GA optimization is used to find optimal parameters for the 15 water balance variables, within the range of parameter bounds provided by the user. Optimization generally takes around 20 minutes. Parameter sensitivity and model accuracy plots are generated and saved.

WB_Model_Accuracy.R

The `WB_Model_Accuracy.R` script evaluates the performance of the water balance model by comparing modeled outputs against observed values for the historical period. It is designed to be run as part of the main `Run_WB_Model.R` workflow and should not be executed independently. The script calculates daily, monthly, and annual aggregations of both actual evapotranspiration (AET) and potential evapotranspiration (PET), comparing model estimates to observations from OpenET. For each variable, it generates scatterplots showing modeled vs. measured values at all three time scales, with trendlines to assess fit (including NSE scores) and time series plots that visualize how modeled and observed values align over time. It's important to note that because the model is calibrated to OpenET estimates of AET, the PET accuracy plots are less informative and should be interpreted with caution.

WB_Future_Analysis.R

The `WB_Future_Analysis.R` script produces water balance projections for the future time period, selects four divergent models to represent a plausible range of climate futures, and plots and analyzes future streamflow trends.

Water Balance Projections

In this script, future water balance projections can be generated in two ways, as defined by the `calcFutureWB` parameter (see User Input Section). The first is through the pre-calculated gridded projections maintained by the NPS and discussed in the Water Balance section. That dataset can be accessed [online](#) via API, and this repository includes a function, `get_conus_wb()`, to pull water balance projections from the gridded product for a single point location. If the online products are not accessible, the data can be accessed by emailing miketercek@yahoo.com and asking for a CSV with projections for a specific location/time period. This repository also includes a function, `get_conus_wb_direct()`, to process water balance data that is provided in CSV format. The second way to generate future projections is to use future meteorological forcing data to run the entire water balance for a given location, using the coefficients that were calibrated on the historical data. The gridded water balance is calculated using coefficients that defined average conditions at each grid cell, taken from other data products (i.e. soil moisture). The calculated water balance uses the coefficients that were calibrated to the historical data; while these might be more accurate for a point location, they also might not necessarily be indicative of physical conditions because they were automatically calibrated. The calculated water balance provides a good option if the gridded water balance data isn't available, or for analyses at point locations. The gridded data is likely more accurate for analyses that span larger spatial extents, or area averages, or analyses where it is important that variables are physically meaningful. If both datasets are available, the code automatically generates a comparison of AET, deficit, and adjusted runoff for a sample year (see below plot).

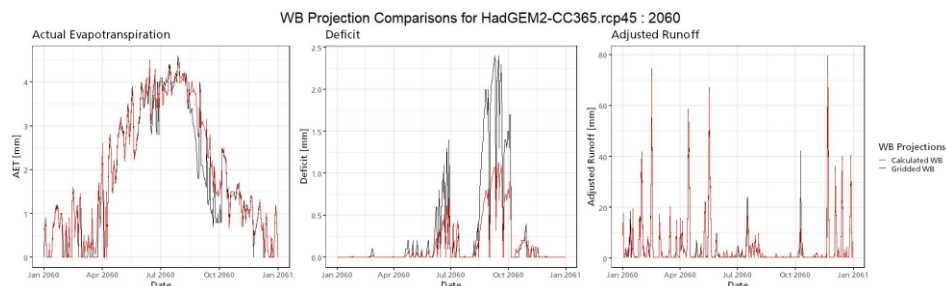


Figure 8. Comparison of gridded and calculated water balance calculations for Little River, TN (USGS #03497300).

Selection of Climate Futures

For plotting and analysis purposes, four divergent climate futures are selected that represent the range of plausible future temperature and precipitation conditions (Lawrence et al., 2021). These climate futures, labelled Warm Wet, Hot Dry, Warm Dry, and Hot Wet, represent the largest possible range of changes in precipitation and temperature by 2050 (calculated as a 30 year average) (Runyon et al., 2024). It is important to remember these names are relative: for example, in the below figure, all futures show precipitation increases compared to the historical period, but the “wet” futures show larger precipitation increases than the “dry” futures. The scenario names can be changed if different names (i.e. Scenario 1/2/3/4) are desired; additionally, a smaller selection (i.e. two out of the four scenarios) can be chosen.

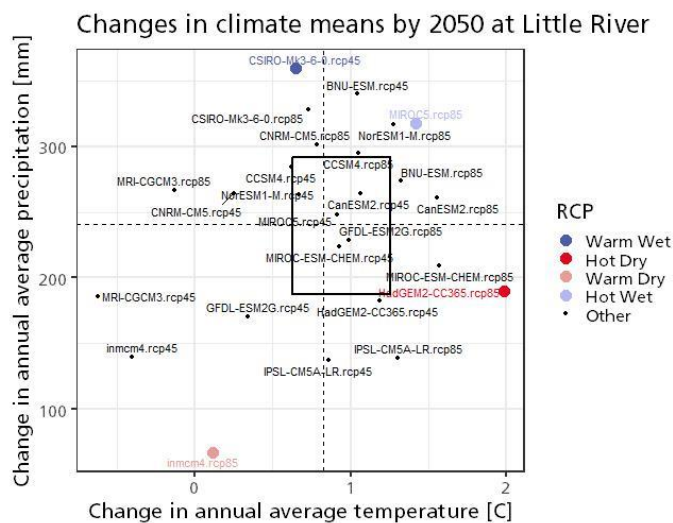


Figure 9. Change in climate means and selected climate futures at Little River, TN (USGS #03497300).

Plots and Analysis

A variety of plots that visualize the future projections are generated by the `WB_Future_Analysis.R` script. These include time series plots, boxplots, heatmaps, scatterplots, ECDF plots, and smoothed temporal averages. In general, these plots highlight the four selected climate futures, though some of the plots show traces from all models to emphasize the intermodel spread between future projections.

The time series plots implement multiple statistical tests to help viewers quantitatively understand trends that may be present in the data. A brief description of each of the statistical trend analyses used in the time series plots is included below.

- LOESS smoothing (locally estimated scatterplot smoothing): This regression technique was used to fit and plot non-linear trend lines. LOESS smoothing is a nonparametric regression technique where polynomials are fitted to localized subsets of the data and combined to generate a smooth trend line. This is included on most of the plots as a red trend line.
- Mann-Kendall test: This nonparametric test evaluates whether statistically significant trends exist in the times series data. Details of the test are discussed in more detail in the **Error! Reference source not found.** section. On the plots, the results of the Mann-Kendall test are included in the top right corner, where the trend is characterized as “Significant” or “Not significant” based on a p-value of 0.05. The p-value of the test is printed below the significance result. Limitations of the Mann-Kendall statistic are described in detail in the Power of Statistical Analyses section.
- Sen’s slope: Sen’s slope (also known as the Theil-Sen Estimator) is a nonparametric method often used in conjunction with the Mann-Kendall test to estimate the slope of a trend in time series data. Sen’s slope calculates the median rate of change between all pairs of data points, producing an estimate that is robust to outliers. On the plots, the calculated Sen’s slope is printed below the Mann-Kendall test results and labeled “Estimated slope.” Multiplying the slope by a given number of years produces the estimated change over that time period.

WB_Multi_Watershed.R

The `WB_Multi_Watershed.R` script enables the user to identify multiple watersheds (defined by name and USGS gage number) and run the streamflow model – including optimization, historical analysis, and future analysis – for all identified watersheds. The script assumes that various model and calibration parameters, such as the PET method or the Boolean variable defining whether the historical analysis script should be run, will remain consistent between all of the watersheds that are being run in the script. These parameters are defined by the user at the beginning of the script. Parameters and the code run in the `WB_Multi_Watershed.R` script are identical to those in the main `Run_WB_Model.R` script and so are not discussed further here. Currently, the script does not have functionality for the user to define their own spatial domain (either through lat/lon coordinates or shapefiles), though it would be quite easy to add this functionality.

```

1 # -----
2 # This script contains code for running and calibrating the streamflow model
3 # across multiple watersheds without additional user-provided input.
4 # -----
5
6 ### Load libraries and function files ###
7 library('here'); lib_install <- FALSE
8 setwd(here('Code')); sapply(list.files(pattern="*.R"), source, .GlobalEnv); setwd(here())
9
10
11 #####
12 #####
13 ### WATERSHED NAMES AND USGS GAGE NUMBERS ###
14 watershed_siteid <- c("Little River", "Cataloochee")
15 watershed_gageid <- c("03497300", "03460000")
16 watershed_foldername <- c('optim', 'optim')
17 num_watersheds <- length(watershed_siteid)

```

Figure 10. The user inputs lists with corresponding watershed names, USGS gage numbers, and folder names for the specific run at the beginning of the code.

WB_Multi_Parameter.R

The `WB_Multi_Parameter.R` script enables the user to manually investigate the impacts of changing model and calibration parameters for a single watershed. After defining the single watershed (including watershed name, corresponding USGS gage number, and spatial domain), the user then inputs configuration and model parameters into a tibble called “v”. Once the first row of the tibble has been defined, the user can copy that row and change individual parameters of interest. This process can be repeated indefinitely, enabling multiple model runs to assess how small parameter adjustments influence calibration accuracy and future projections. Otherwise, the code and parameters in the `WB_Multi_Parameter.R` script is identical to the code in the main `Run_WB_Model.R` script and so are not discussed further here.

```

51 # Store all parameters in tibble
52 v = tibble(
53   # Folder name
54   FolderName = "optim_Oudin",
55
56   # General model variables
57   optimization = TRUE, optimization_var='AET', point_location = TRUE, percent_skill_cutoff = 0.1,
58
59   # Meteorological variables
60   GridMET = TRUE, incompleteMonths = FALSE, fillLeapDays = TRUE, delayStart = FALSE,
61   tmmx_slope = 1, tmmx_bias = 0, tmmn_slope = 1, tmmn_bias = 0, p_slope = 1, p_bias = 0,
62
63   # Water balance variables
64   PETMethod = "Oudin", calcFutureWB = TRUE, userSetJTemp = FALSE,
65   gw_add=0, vfm = 0.7555, jtemp = 1.982841, jrange = 3, hock = 4, hockros = 4,
66   dro = 0, mondros = 0, aspect = 180, slope= 0, shade.coeff= 1, SWC.Max = 200,
67   k_c = 1, et_slope = 1, et_bias = 0,
68   Soil.Init = SWC.Max, Snowpack.Init = 0, T.Base = 0,
69 )
70
71 ### Create additional rows of variables and change as desired ###
72 v[2,] <- v[1,]; v[3,] <- v[1,]
73 v[2,"PETMethod"] = "Penman"; v[2,"FolderName"] = "optim_Penman"
74 v[3,"PETMethod"] = "Hamon"; v[3,"FolderName"] = "optim_Hamon"

```

Figure 11. In this example code, the user investigates the impacts of PET method on model accuracy by holding varying the PET method and re-running the model while holding all other variables constant. By changing the folder name with each run, the user ensures that the output from each run is stored in a different location.

Streamflow Scripts

The Streamflow folder includes scripts to calibrate and run a model of streamflow in reference to a user-provided USGS stream gage. These scripts are organized in the main `Run_Streamflow_Model.R` script and are primarily sub-scripts that need to be called from the main script, except for the `Streamflow_Historical_Analysis.R` script that can be run independently. The structure and components of the main `Run_Streamflow_Model.R` script, in addition to those of the supporting scripts, are described in the following section.

Run_Streamflow_Model.R

The script has four parts: defining model parameters and setting initial values of variables, pulling streamflow observations and meteorological forcing data, running the model (including optimization, if desired), and analyzing model results.

Both the water balance and streamflow components of this model require parameterization; the water balance model includes 12 parameters defining various physical characteristics of the watershed, while the IHACRES model includes 5 parameters describing the characteristics of the streamflow. This model calibrates both sets of parameters for the historical time period, using recorded observations of streamflow from a given USGS gage. The initial version of the model existed in spreadsheet form; users could manually adjust each coefficient and evaluate the model's accuracy through calculated NSE values and plots. Though this code can be run with provided coefficients, such as those produced by a calibrated spreadsheet version of the model, this R code automates the process of finding the optimal coefficients for both components of the model.

User Input

The top of the `Run_Streamflow_Model.R` script includes a section that requires user input, titled "Set user-defined variables." User-provided parameters are identical to those in the `Run_WB_Model.R` script, described above, with the addition of the following variables for the streamflow component of the model:

- `historical_analysis`: TRUE runs the `Historical_Analysis.R` script to analyze historical streamflow observations for the identified USGS gage.

Streamflow variables:

- `delayStart`: TRUE to delay starting year by 11 years from the user-defined value, allowing the initial parameters of the streamflow model to calibrate. FALSE to use the user-defined value as starting year.
- `NonZeroDrainInitCoeff`: TRUE to use nonzero initial drainage coefficients that are the average of the modeled historic January Flows after the Cutoff Year. FALSE to set initial drainage coefficients to 0. Typically FALSE.
- `flow_components`: change the number of components that characterize the IHACRES flow. can be 2 or 3. 2: flow has quick and slow components; 3: flow has quick, slow, and very slow components.

Optional Parameters:

- Default model parameters: the streamflow model includes additional parameters for the IHACRES model in addition to the water balance parameters described in the `Run_WB_Model.R` section.

- Model parameter optimization limits: the streamflow model also includes additional parameter optimization limits for the IHACRES model.

Streamflow_Optimization.R

Optimization occurs in two steps. First, the optimal water balance variables are identified, using monthly NSE as the variable to optimize the parameters on. Because the water balance variables describe general characteristics of the watershed, they primarily impact streamflow over a longer time scale than daily. Additionally, using a monthly time scale increases the efficiency of the water balance optimization process, which is already complex because it involves 12 variables. A GA optimization function is used to find optimal parameter values for the water balance section of the model. Second, the optimization of IHACRES model parameters involves two optimizations. First, a GA optimization function is used to find the approximate values of the parameters. Second, `optim()` is used to search the parameter space near the approximate parameter values and find the most optimal parameters. In the test case, the adoption of the second optimization increased the NSE by 0.02. `optim()` is not used alone because it tends to only search locally, which can produce incorrect values if the initial values provided to the function are incorrect and does not allow for exploration of the parameter space. The optimization of the IHACRES model parameters uses the daily NSE as the optimization value. The IHACRES parameters contribute significantly to streamflow at a daily scale and tend to be more sensitive than the water balance variables, which is why two optimization functions are used for the IHACRES variables. Overall, optimization takes about 30 minutes.

Commented [MB2]: How long?

Streamflow_Model_Accuracy.R

The `Streamflow_Model_Accuracy.R` script provides visualizations of model fit and accuracy during the historical period of record for which both measured and modeled data is available. If using GridMET or Daymet meteorological data, this period will span 1979 or 1980 (respectively) to present day *or* the USGS streamgage period of record, depending on which is shorter. This script cannot be run independently; it must be called from `Run_Streamflow_Model.R`.

Analysis plots include scatterplots and time series of measured and modeled data, for daily, monthly, and annual aggregations. The script also produces a quantile plot, which shows linear regression lines and pseudo R^2 values for the 1st, 10th, 25th, 50th, 75th, 90th, and 99th percentiles of a scatterplot of measured vs modeled data. The script also generates Empirical Cumulative Distribution Function (ECDF) plots and compares the measured and modeled distributions using a Kolmogorov–Smirnov test (Popat & Hartmann, 2024). These plots are saved within the model run sub-folder, as described in the **Repository Overview** section.

Streamflow_Historical_Analysis.R

The `Streamflow_Historical_Analysis.R` script provides visualizations and preliminary analyses of historical measured streamflow trends and can be run independently or in conjunction with the `Run_Streamflow_Model.R` script. The `Streamflow_Historical_Analysis.R` script is based on the assumption that historical streamflow measurements have a low level of anthropogenic influence; the visualizations and trend analyses do not take into account any disruptions or changes due to anthropogenic changes, so streams with little to no anthropogenic influence will result in more accurate analyses. To determine whether a specific stream has experienced anthropogenic impacts and therefore would not be a good fit for this analysis, the USGS GAGES-II dataset identifies approximately 2000 least-disturbed “reference streams” across the United States (Falcone, 2011). Additionally, plots generated by the `Streamflow_Historical_Analysis.R` script, including plots of annual volume and heatmaps of daily streamflow, can be visually inspected to determine whether the stream has experienced

significant anthropogenic changes (Strandhagen et al., 2006). In the `Run_Streamflow_Model.R` script, a Boolean variable `historical_analysis` will call the `Streamflow_Historical_Analysis.R` script when TRUE, using the user-provided site name and USGS gage number to pull historical streamflow data. When the `Streamflow_Historical_Analysis.R` script is run independently, the user will need to provide a USGS gage number and site name to run the code.

Given a USGS gage number and SiteID, the `Streamflow_Historical_Analysis.R` script pulls measured streamflow data from gage for the entire historical period of record and saves the data as a CSV file, allowing the user to easily access the data for further analysis. The script generates various plots and analysis metrics to evaluate historical trends in the data. A range of metrics were assessed to quantify the hydrologic character of the stream (Al-Chokhachy et al., 2017; Dhungel et al., 2016), including:

- Annual volume
- High flow (number of days above historical 95th percentile)
- Low flow (number of days below historical 5th percentile)
- Q7 Min (average 7 day minimum streamflow)
- Q7 Max (average 7 day maximum streamflow)
- Timing (50% flow date)
- Streamflow efficiency (ratio of streamflow to precipitation)
- Customized metrics (specific flow volumes or time periods relevant to stream characteristics or species of interest)

These metrics are calculated and plotted using data from the historical USGS streamgage record, and a variety of statistical techniques are used to investigate the magnitude, direction, and consistency of trends in these metrics. These methods are described briefly below. Not all plots include all of these analyses; for most metrics, the code can be adjusted to change the analysis and/or whether it is shown on the plot. Descriptions for the metrics not included here can be found in the `WB_Future_Analysis.R` section.

- LOESS smoothing
- Mann-Kendall test
- Sen's slope
- Pettitt test: The Pettitt test is a nonparametric statistical test used to detect a single change point in a time series of data by comparing the distributions before and after each possible change point. It assumes there is at most one change point that marks an abrupt shift in the data, so it may not capture more gradual trends or trends with multiple change points. The change point identified by the Pettitt test is included in some of the plots as a vertical red dashed line.

[Streamflow_Future_Analysis.R](#)

The `Streamflow_Future_Analysis.R` script produces water balance projections for the future time period, generates projections of future streamflow using the calibrated IHACRES coefficients, selects four divergent models to represent a plausible range of climate futures, and plots and analyzes future streamflow trends. A more detailed description of each of these steps is included in the `WB_Future_Analysis.R` script. The scripts are similar, except the `Streamflow_Future_Analysis.R` contains an additional step in which future water balance projections are run through the IHACRES streamflow model to generate projections of future streamflow. The scripts also output slightly different plots to highlight distinct water balance and streamflow trends.

[Streamflow_Multi_Watershed.R](#)

This script is identical to the `WB_Multi_Watershed.R` script, but it runs the coupled water balance – IHACRES flow model rather than only the water balance model. For a more complete description, see documentation for the `WB_Multi_Watershed.R` script.

[Streamflow_Multi_Parameter.R](#)

This script is identical to the `WB_Multi_Parameter.R` script, but it runs the coupled water balance – IHACRES flow model rather than only the water balance model. For a more complete description, see documentation for the `WB_Multi_Parameter.R` script.

Supporting Code

The Code folder includes several scripts that define functions for other scripts in the repository, including `DataFunctions.R`, `WBFunctions.R`, and `ModelFunctions.R`. These scripts house supporting code for the Water Balance and Streamflow models. Additionally, this folder contains a Jupyter Notebook titled `get_climate_data.ipynb` that contains code to pull meteorological data for the water balance and streamflow models. This script is faster and more reliable than the corresponding R code to pull meteorological data in the `DataFunctions.R` script.

[DataFunctions.R](#)

The `DataFunctions.R` script contains a variety of functions that access, clean, and process various types of data that are used in the water balance and streamflow models. Data is available at both point locations and as area averages for GridMET, Daymet, MACA historical, MACA future, and OpenET ET and ETo products. Additionally, point location data from a single USGS streamgage is available. Data is available only at a point location for the gridded NPS water balance data maintained by Mike Tercek (and that data may be inconsistently available, depending on the status of the server the data is stored on). The script also contains functions to pull and process spatial watershed data for a given USGS gage and identifying four divergent climate futures based on future temperature and precipitation projections.

[WBFunctions.R](#)

The `WBFunctions.R` script provides a comprehensive set of functions for calculating intermediary and final variables in the water balance model. The script includes functions for each step in the water balance model, including precipitation partitioning into rain and snow using a temperature-based freeze factor, water availability at the soil surface, soil water content, climatic water deficit, direct and total runoff, growing degree days, and snowmelt and snowpack accumulation. The script supports multiple methods for estimating PET, including Hamon, Thornthwaite, Oudin, and Penman-Monteith, with optional adjustments for heat load, shade, and bias correction. AET is computed based on PET, with constraints to prevent exceedance of PET. Finally, the script also includes supporting functions used in the modeling of these processes, including bias adjustment, vapor pressure, relative humidity, and radiation balances. The functions in the script are a combination of the [WaterBalance R package](#) and other functions written explicitly for this model.

[ModelFunctions.R](#)

The `ModelFunctions.R` script includes functions for running the water balance and IHACRES streamflow components of the models, as well as functions facilitating the optimization calculations of both models. `WB()` runs the water balance model and `Drain()` runs the IHACRES flow model. `WB_Optim()` and `IHACRES_optim()` are the optimization functions for the combined water

balance/streamflow model, and `WB_Optim_ET()` is the optimization function for the water balance model. The script contains additional supporting functions, including functions to calculate initial model coefficients for the IHACRES streamflow model and aggregate monthly flow projections.

Validation

The Validation folder includes scripts and supporting files to run the water balance model and confirm that its output matches the original version of the model in Excel, using the test site of Frog Rock in Yellowstone National Park. Model outputs are also compared to the Python model, an adaptation of the original Excel model into Python, to ensure that all three versions of the model accurately reproduce the core functionalities of the original NPS water balance model.

The script `FrogRock_crosscheck.R` contains code to run the R version of the water balance model and compare it to saved results of the model run in Python and Excel (these results are stored as CSV files in the Validation folder). The code also generates figures comparing the R code to both Python and Excel; these figures are stored in the Output/Validation folder.

Development Notes

Point Location vs. Area Average

In topographically complex regions, the spatial scale at which meteorological data is accessed can alter the magnitude and even direction of projected climatological changes. The repository contains functions to pull data for a point location or as an area average. By default, the point location is the centroid of the watershed and the area corresponds to a watershed area, but users can also input their own spatial domains. As the figures below show, selection of point location or area average spatial scales can alter projected climatological changes, so careful selection of spatial scale is important.

In general, area averaged data tends to smooth out extremes and may provide a more accurate representation of streamflow across a larger spatial domain, as it takes into account the cumulative precipitation across a watershed, rather than merely one grid cell. In contrast, point location data is likely more accurate for small-scale or localized analyses, topographically complex watersheds, and/or locations characterized by flashy or localized precipitation (such as the western U.S.)

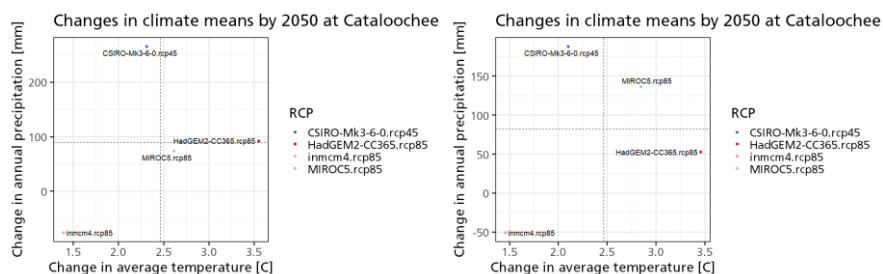


Figure 12. Area averaged climate data is on the left and point location climate data is on the right, for the Cataloochee Creek watershed (USGS gage number 03460000). Both compare future MACA data to

historical MACA data, but the magnitudes of precipitation vary based on the spatial scale, especially for the MIROC5 (RCP 8.5) model.

High Flow Events

The streamflow model is known to perform poorly in high flow events on all watersheds, regardless of their location or climate characteristics. Several reasons contribute to this. First, several variables contribute to flood intensity, but the model doesn't include all of these variables. In the list below, check marks indicate variables that the model accounts for.

- How quickly the storm is moving
- How intense the rainfall rates are
- Whether storms are re-developing and repeatedly impacting the same area
- ✓ How much rainfall becomes surface runoff and where it drains to
- How porous the soil is
- ✓ How much water the soil already holds
- Amount and type of vegetation covering the soil
- ✓ How much of the land surface is paved and won't hold water
- Whether there are storm drains to effectively convey the water out of the area
- How steep the terrain is
- ✓ Whether the land has been impacted by recent wildfire
- ✓ Depth of snowpack
- Condition of snowpack

Therefore, the model structure inherently limits its accuracy in resolving high flow events. Additionally, the model relies on gridded meteorological data, which is provided at a daily scale (whereas many extreme precipitation events occur over shorter time scales) and may be less accurate due to its gridded nature. Overall, despite several attempts to improve model performance, it remains limited in its accuracy in modeling peak flow events and likely will into the future, barring significant changes to its structure.

However, the model can still provide useful information about peak flow events. The model is highly accurate at matching the timing of peak flow events. One potential method for using the model to evaluate peak flow projections includes comparing future projections to *modeled* historical projections, rather than measured observations. Given the model cannot capture the magnitude of peak flow events during either the historical or future period, this comparison is an “apples-to-apples” way to evaluate changes in the relative frequency, intensity, or magnitude of peak flow events.

Power of Statistical Analyses

The primary trend analysis test used in this analysis was the Mann-Kendall test, which is a non-parametric statistical test used to detect monotonic (strictly increasing or decreasing) trends in time series data. The test was run for the entire time series, rather than a rolling window, because it requires large amounts of data points for accurate comparison. Statistical significance was determined with a p-value of 0.05. This value is quite low, meaning the trends in the data must be very robust and strictly monotonic in order to be classified as significant by the Mann-Kendall test. Thus, while the Mann-Kendall test may not classify a given trend as “significant” by statistical measures, the trend may still exist and be impactful enough that it should be considered in planning or management decisions. On the other hand, the Mann-Kendall test is more likely to produce statistically significant trends on larger datasets, even if the change is small, because the test statistic depends quadratically on the number of pairwise comparisons in the

data. In the above analyses, the future analyses produced more statistically significant test results than the historical analyses, because the future time periods were longer and included more datapoints. Evaluating the Sen's slope and visual trend in the data is equally as important as considering the Mann-Kendall test in decision-making.

The Mann-Kendall test is also limited because it is run for the entire time series, rather than a rolling window. This means that trends must be consistent and monotonic across the entire period of record, can span up to 100 years for USGS gage records. This may not capture interannual variability, seasonal trends, or non-monotonic trends. To evaluate whether trends were occurring on smaller time scales, the Mann-Kendall test was run on a rolling time window at both a daily time scale (with a rolling window of three years) and an annual time scale (with a rolling window of ten years) for both rivers. A plot of Mann-Kendall p-values with a daily rolling time window of three years is shown below; though there are periods of statistically significant trends, such as during the early 2000s, these trends are not robust enough to suggest robust shifts in Mann-Kendall test results during periods of the observational record.

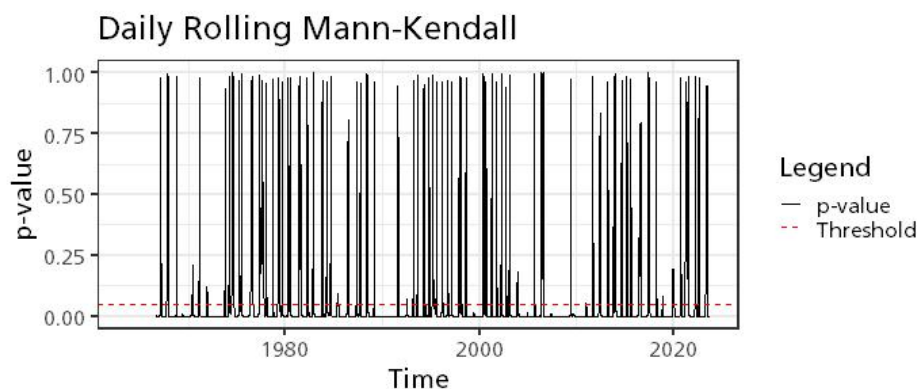


Figure 13. Daily rolling Mann-Kendall test using a three year time window for Little River, TN (USGS #03497300).

Known Weaknesses

Calibration of Water Balance Model

The calibration of the water balance model to OpenET data relies on forcibly adjusting the magnitudes of PET and AET, causing the model to veer from physical reality. With parameters calibrated to the streamflow model, the water balance model generally appeared to underestimate AET when compared to OpenET observations. Matching the magnitude of the OpenET data required shifting the modeled PET data using the crop coefficient as a multiplier and shifting the modeled AET data using both a bias and a slope adjustment. Because of these adjustments, the outputs of the water balance calibration may not appear physically realistic and/or the water balance may not close. It is important to note that OpenET data may not be the best source for ET calibration – the discrepancies between the modeled ET and OpenET data could be due to differences in the algorithms that OpenET uses to estimate ET from satellite imagery. Ideally, this calibration should occur using more directly measured ET data, such as an observational network like FLUXNET, but that data is only available for point locations. Overall, more

work on the ET calibration is required before a model calibrated to ET should be used for analysis. It is recommended instead to use a streamflow-calibrated water balance model or a manual spreadsheet calibration.

Data Sources

The repository currently relies on the ClimateR package to pull historical and future meteorological data from GridMET, Daymet, and MACA v2 downscaled projections. Unfortunately, I have had many technical difficulties with the ClimateR package. Data pulling methods are slow and prone to interruptions, the format of the results seems to constantly change, and it does not appear that the package is actively maintained. To the best of my knowledge, the functions that I've written access and process the data accurately, but I was unable to comprehensively test them before my term ended due to technical difficulties that I continued to encounter with the ClimateR package. Therefore, I recommend that users find alternative methods to download climate data, rather than relying on the scripts in the R repository. There are two main approaches: first, code to access GridMET and MACA data is available in a Jupyter Notebook in the Code folder titled `get_climate_data.ipynb`; and second, users can download their own offline copy of MACA/GridMET data to access (other CCRP members who have worked extensively with MACA data have adopted this approach). If users want to use the functions that I have written, they should thoroughly test them in comparison to output from the Jupyter Notebook.

OpenET data accessed through the API has also been inconsistently available. To access the data, users must register an account and get a key to access the API (more information found [here](#)), and this key must be defined in the `DataFunctions.R` script. The [OpenET website](#) states that monthly data is available from 2000 – 2025 and daily data is available from 2016 – 2025, but the API does not consistently allow data from that time period to be pulled. These years are set as the default years for pulling data from, but if the data request fails, smaller temporal periods will be tried until the data can be pulled. This may be a result of ongoing data maintenance by OpenET, or it may represent a change in their policies – it is unclear.

Finally, the code doesn't integrate well with the gridded water balance product generated and maintained by Mike Tercek. There is some functionality for pulling gridded water balance data at a point location, but no functionality for pulling area-averaged gridded water balance data. Given that the water balance data is often not publicly available due to high server costs, it doesn't make sense to build functions that won't work when the server is offline. However, this means that users who want to analyze the gridded water balance data must directly contact Mike Tercek and wait for him to send data, which is inefficient. Finding a way to publicly host the gridded water balance data (and building access to that into the data processing/analysis pipeline) would be a great way to expand the capabilities of the model and align with previously completed water balance projects.

When this model was developed, GridMET and MACA v2 downscaled CMIP5 data were the most accurate, consistently available sources of meteorological data for the historical period and the future period, respectively. As more downscaled data products become available for CMIP6, it is likely that MACA will no longer be the gold standard for future projections, and the model will need to be adapted to include new data sources.

Areas for Continued Development

There are many opportunities to continue to expand the capabilities and functionalities of the water balance and coupled water balance – IHACRES streamflow model. This section summarizes a few of the opportunities that have been discussed (though this list only scratches the surface of the development possibilities!)

- Include leaf area index (LAI): previous versions of the water balance model have included calibration with satellite observations of LAI. In addition to serving as a good way to accurately calibrate evapotranspiration estimates, including LAI allows for investigation of the impacts of changing land cover on hydrology.
- More advanced WB functions (snowmelt function): various functions in the water balance model can be modeled in more complex ways. For example, the current snowmelt function calculates melt as a relatively simple combination of rainfall-induced and temperature-driven melt; implementing a more complex snowmelt function, such as the one included in the HBV model, could improve accuracy in snowmelt-driven watersheds (Bergström, 2015).
- Integration with NPS gridded water balance product: because the NPS gridded water balance data isn't consistently stored on a publicly available server, there are currently limited opportunities to integrate the R version of the model with the gridded product. Integration would lead to interesting and useful comparisons (especially related to model accuracy with various different parameter choices). Developers interested in integration should contact Mike Tercek (miketercek@yahoo.com).

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